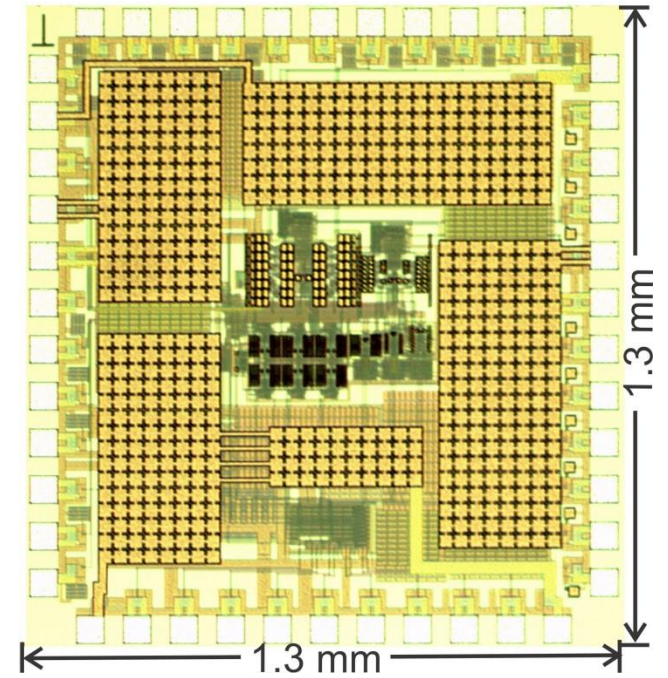
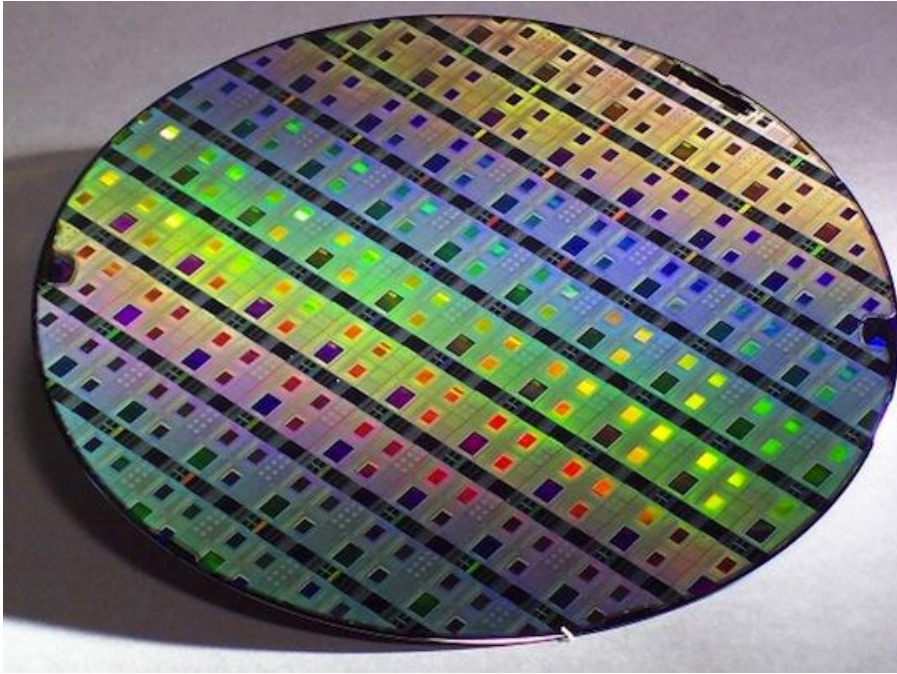


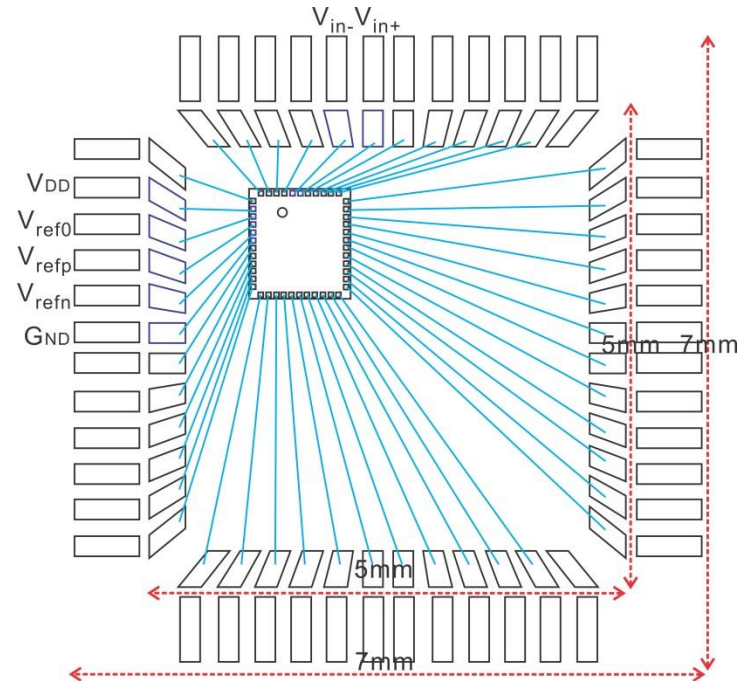
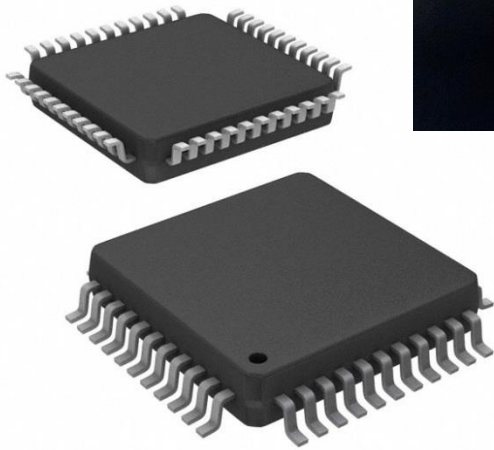
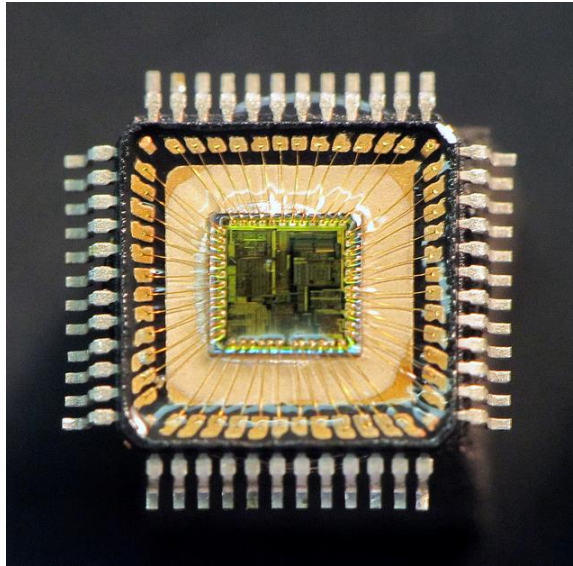
Data Converter Testing

Nan Sun
Tsinghua University
nansun@Tsinghua.edu.cn

Get the Die

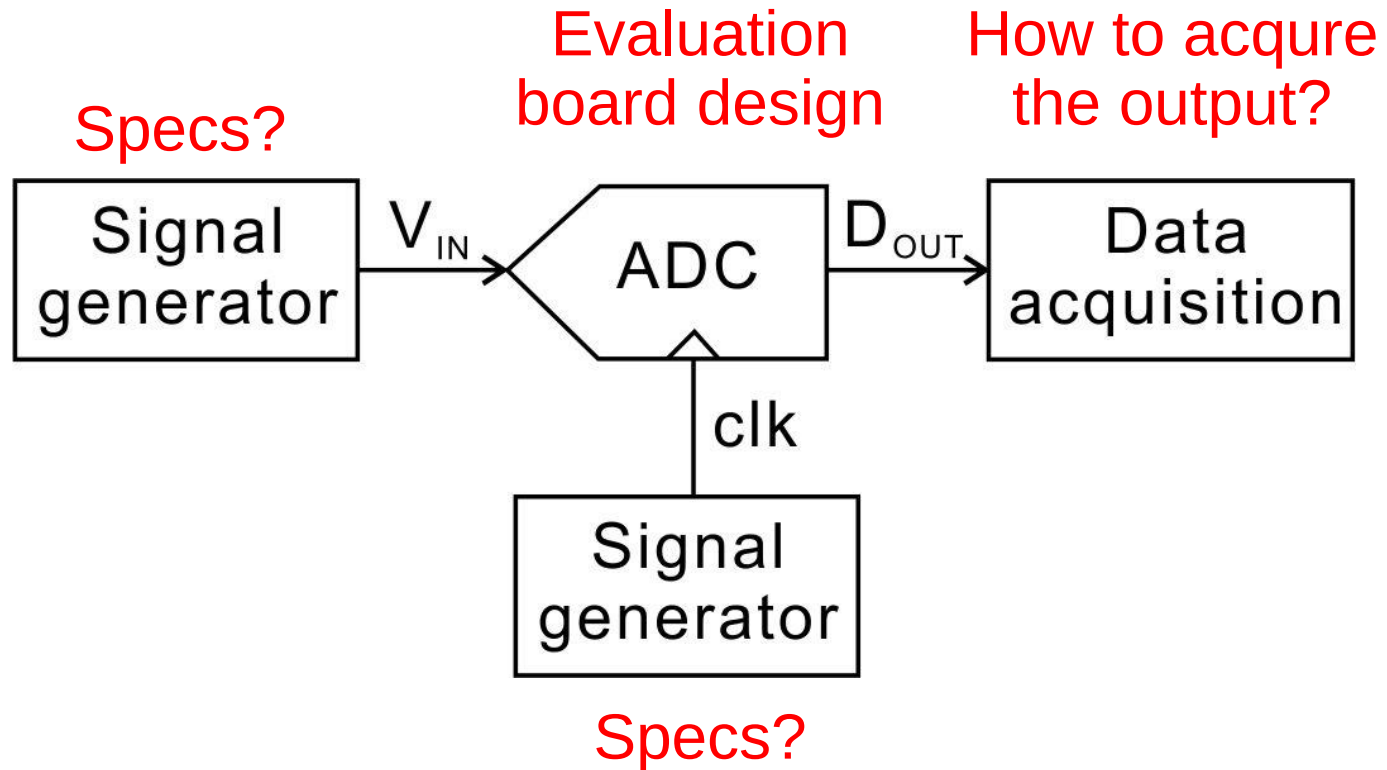


Packaging



Off-centered
packaging

ADC Test Setup



High-Performance ADC Specs

Resolution	> 14b
Conversion rate	> 100 MS/s
Input range	2V peak-to-peak
SNR	> 80 dB
SFDR	> 90 dB
DNL	< 1 LSB
INL	< 1 LSB

- Testing a high performance ADC is very challenging!
 - Almost as hard as designing one
 - Test instruments need to be much better than the ADC under test
- We need to be aware of the test setup and the equipment limitations

A Low-Cost Signal Source

- We want the input signal to be very linear
 - E.g., SFDR > 90dB @ 100 MHz
- How about using a low-cost signal generator?
 - Frequency range limited to 15MHz
 - Large harmonic distortion: -40dBc for $f_{\text{in}} > 1\text{MHz}$
 - Far away from what we need!

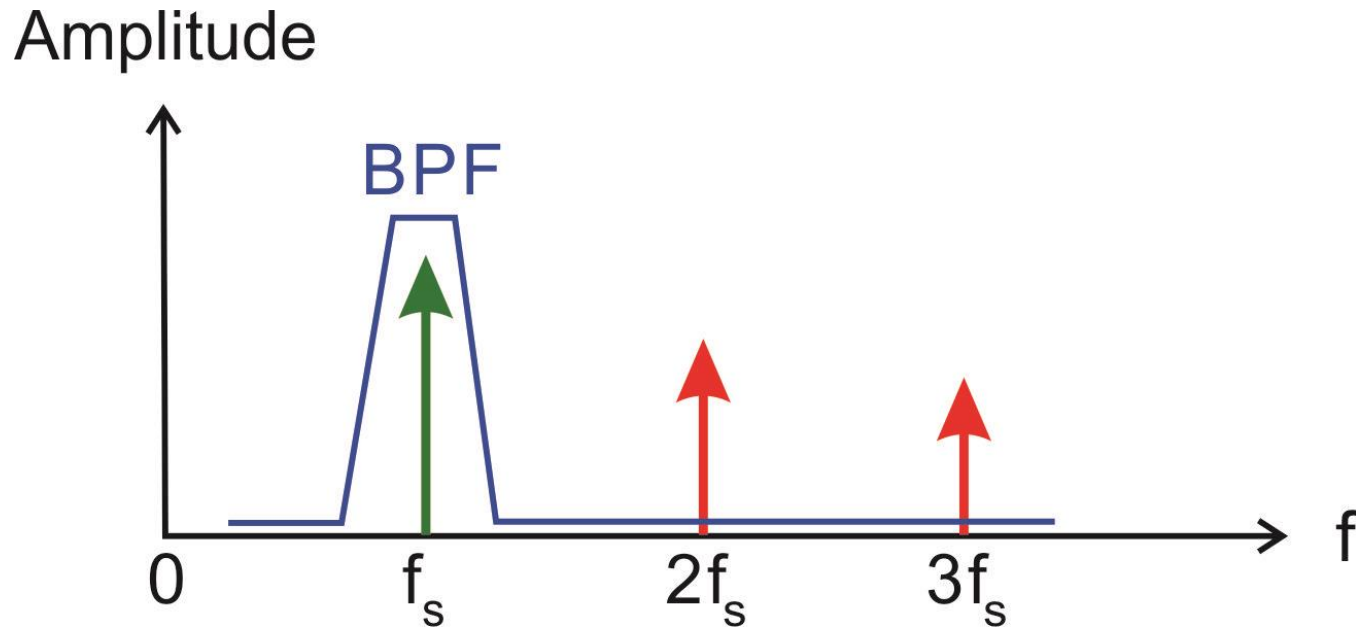


An Expensive Signal Source

- Now let us use an expensive (\$50K) signal generator
- Frequency range is significantly extended
 - Up to 20GHz
- But distortion problem is still not solved
 - Large harmonic distortion: -40dBc
 - How to remove the distortion?



Filtering Out Harmonics



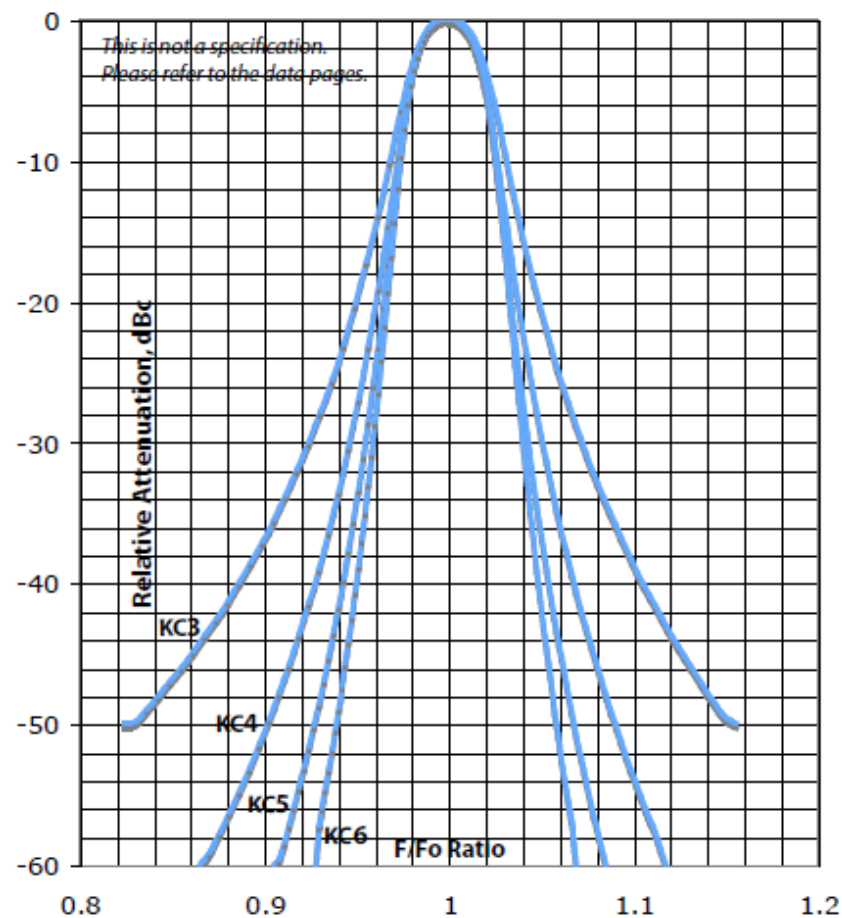
- Given $HD = -40\text{dBc}$, we need a stopband rejection $> 50\text{dB}$ to ensure $SFDR > 90\text{dB}$
 - The design of the filter is nontrivial

High-End Filters



www.tte.com

www.allenavionics.com



Tunable Filters



Manually tune



Digitally tune

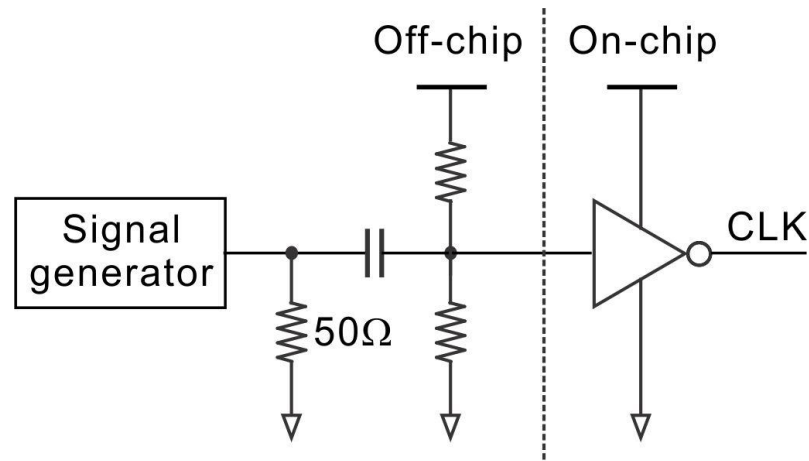
www.klmicrowave.com

Filter Distortion

- Filters themselves generates distortions!
 - Many filters are designed to operate with a small signal swing
 - In ADC testing, the input swing is usually large $\sim 2V$
 - Low-cost filters may not satisfy the linearity requirement
 - The self-generated distortion may be even larger
- Filter distortion is usually not specified
 - Need to call manufacturer
 - High-end filters usually have $HD < -85\text{dBc}$, but don't trust your filters blindly
 - Use a spectrum analyzer to test the filter distortion before using it
 - The spectrum analyzer itself generates distortion!
 - We can place attenuators in front of the spectrum analyzer

Clock Generator

- The clock source needs to have a high quality
 - We do not care about distortion
 - We care about phase noise or jitter
- High frequency clocks are always sine wave like
 - We need a clock receiver to turn it into a square wave
 - Be careful about the phase noise of the clock interface circuits
 - Ideally, the jitter should be limited by the clock source



Phase Noise and Jitter

- For a clock generator, its phase is not fixed, but follows a random walk

$$\langle \phi_n^2 \rangle = 2Dt$$

- Where D is the phase diffusion constant
- Low-end signal generator:
 - Large D, cycle-to-cycle jitter $\approx 100\text{ps rms}$
- High-end signal generator:
 - Small D, cycle-to-cycle jitter $\approx 0.1\text{ps rms}$

Jitter from Other Circuits

- Other clock interface circuits also contribute jitter.
- Clock jitter from other circuits can be minimized by:
 - Separate supplies
 - Separate analog and digital clocks
 - Short inverter chains between the clock source and the destination
- Jitter induced noise has the following unique properties
 - RMS noise proportional to input frequency
 - RMS noise proportional to input amplitude
- Thus, it is easy to identify when ADC noise is jitter limited

Jitter Estimation

- Jitter can be measured even if the ADC is not jitter limited
 - D.M. Hummels, W. Ahmed, W., F.H. Irons, "Measurement of random sample time jitter for ADCs," *Proc. ISCAS*, pp.708-711, May 1995.

$$\begin{aligned}
 x(t) &= A \cos(\omega_0 t + \theta) \\
 y_k &= A \cos(\omega_0 k T_s + \theta) + g(x(t))|_{t=kT_s} \\
 &\quad - A \omega_0 \Delta_k \sin(\omega_0 k T_s + \theta) + n_k
 \end{aligned}$$

After removing all harmonics:

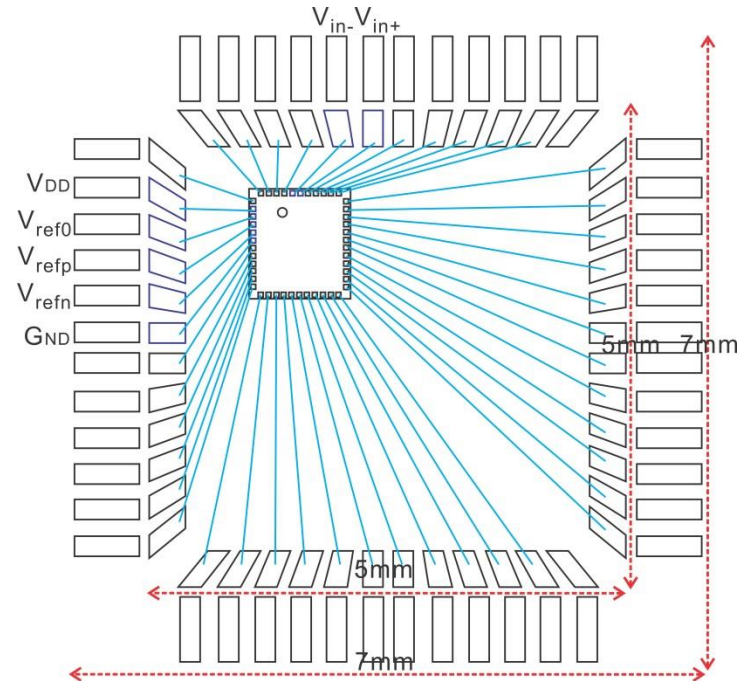
$$e_k = -A \omega_0 \Delta_k \sin(\omega_0 k T_s + \theta) + n_k$$

$$\begin{aligned}
 E\{e_k^2\} &= E\{A^2 \omega_0^2 \Delta_k^2 \sin^2(\omega_0 k T_s + \theta) + n_k^2\} \\
 &= E\left\{\left(\frac{A^2 \omega_0^2 \Delta_k^2}{2} + n_k^2\right) - \frac{A^2 \omega_0^2 \Delta_k^2}{2} \cos(2\omega_0 k T_s + 2\theta)\right\} \\
 &= \left(\frac{A^2 \omega_0^2 \sigma_\Delta^2}{2} + \sigma_n^2\right) - \frac{A^2 \omega_0^2 \sigma_\Delta^2}{2} \cos(2\omega_0 k T_s + 2\theta)
 \end{aligned}$$

This is a component at $2\omega_0$ that is proportional to jitter σ_Δ

Evaluation Board

- The ADC pin-out need to be carefully planned
 - We need to place sensitive analog pins far away from “dirty” digital output pins
- Key considerations in the board design
 - Supply/ground routing
 - Bypass capacitors
 - Coupling between signals
- Check out ADC vendor datasheets for example schematics and layouts

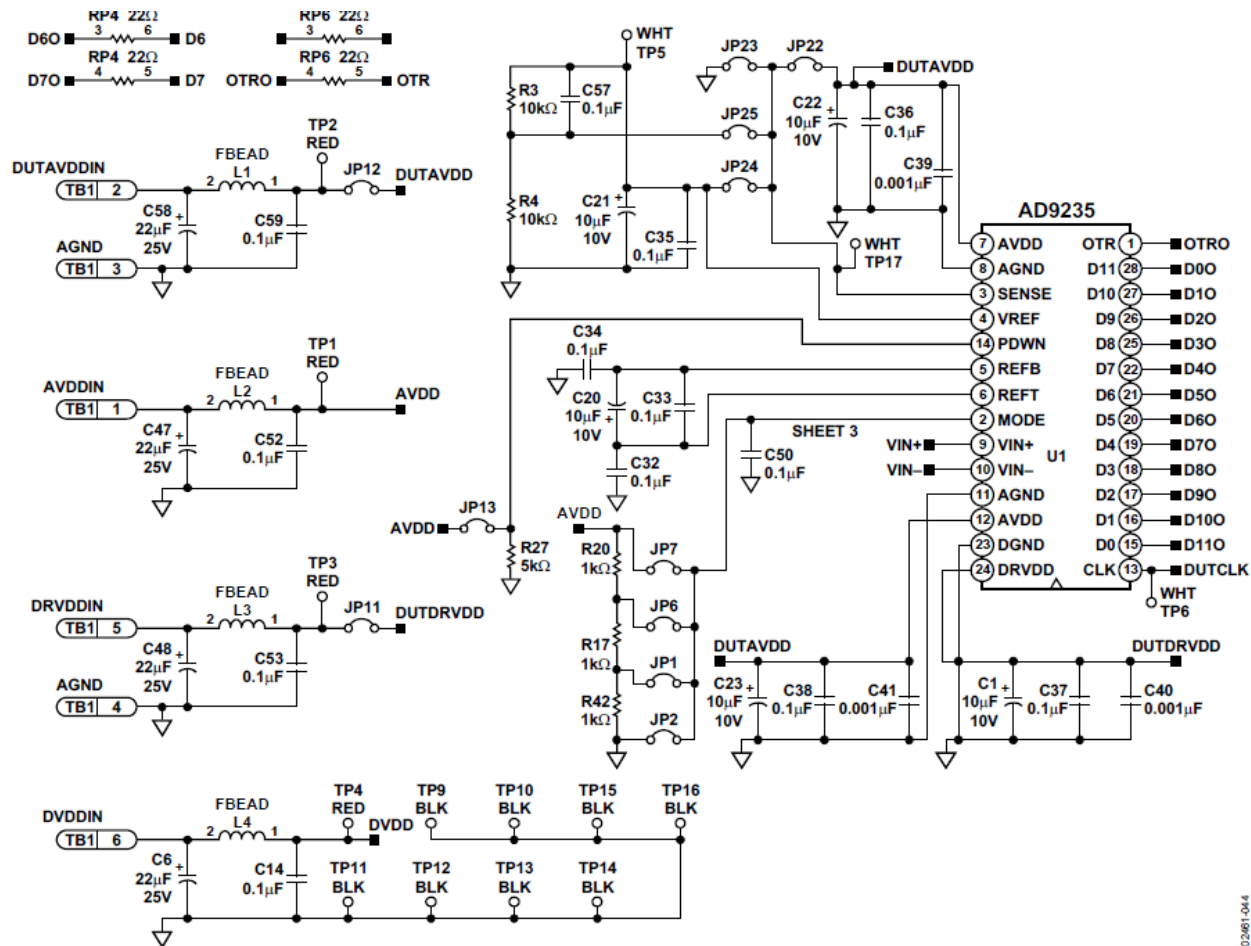


Good References on Board Design

- Analog Devices Application Note 345: "Grounding for Low-and-High-Frequency Circuits"
- Maxim Application Note 729: "Dynamic Testing of High-Speed ADCs, Part 2"

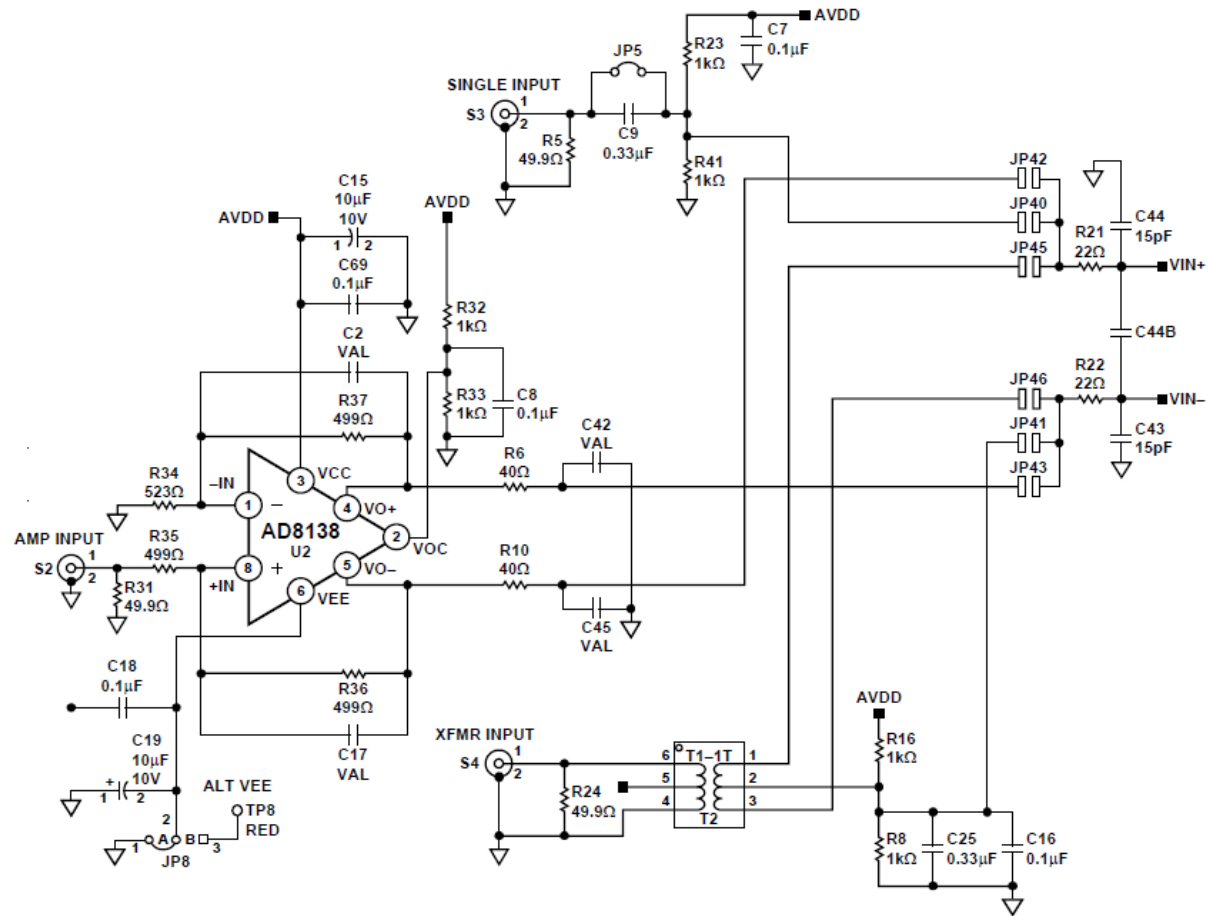
Example Board Schematic (1)

[Analog Devices AD9235 Data Sheet]

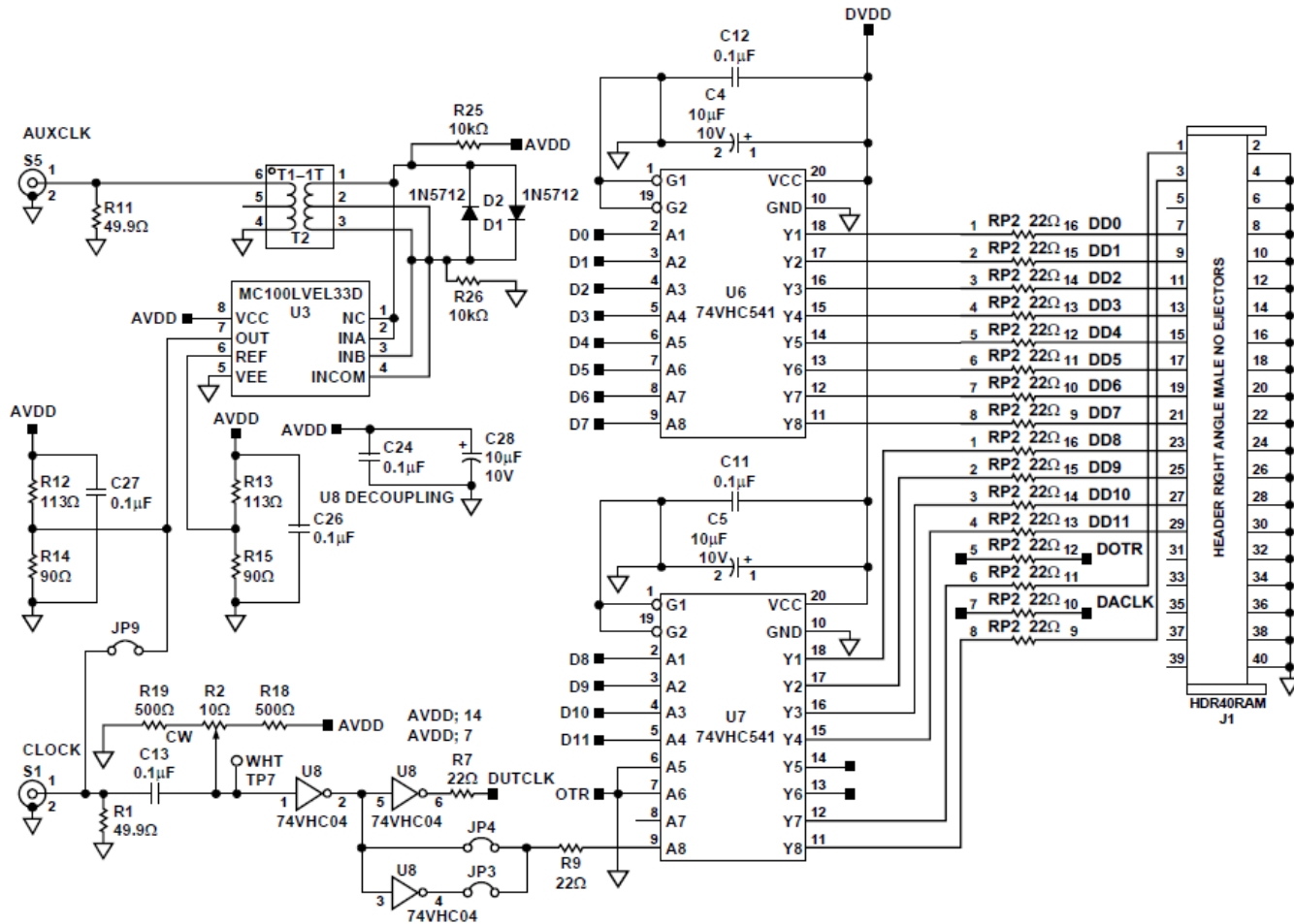


02481-044

Example Board Schematic (2)



Example Board Schematic (3)



Example Vendor Board Layout

[Analog Devices AD9235 Data Sheet]

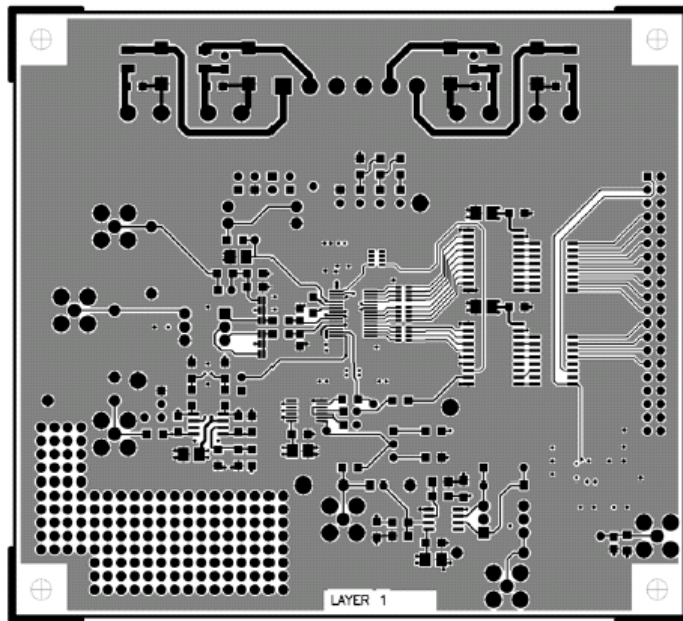


Figure 21. TSSOP Evaluation Board Layout, Primary Side

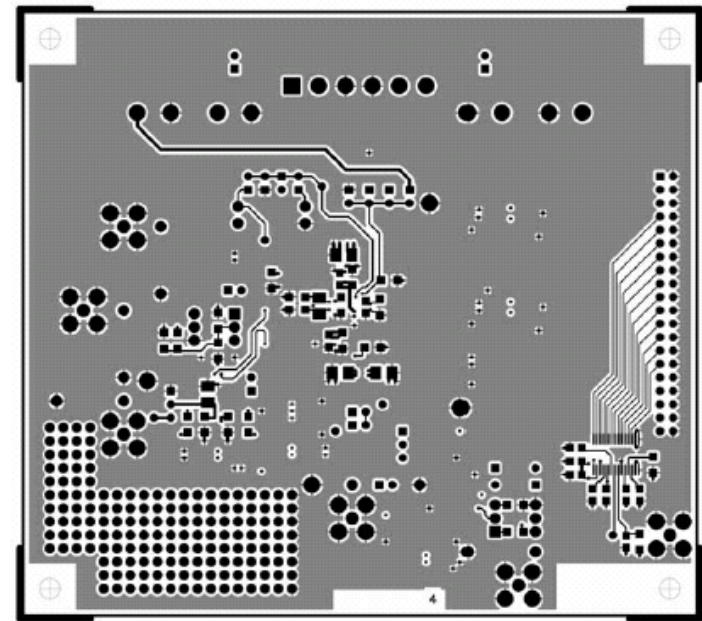
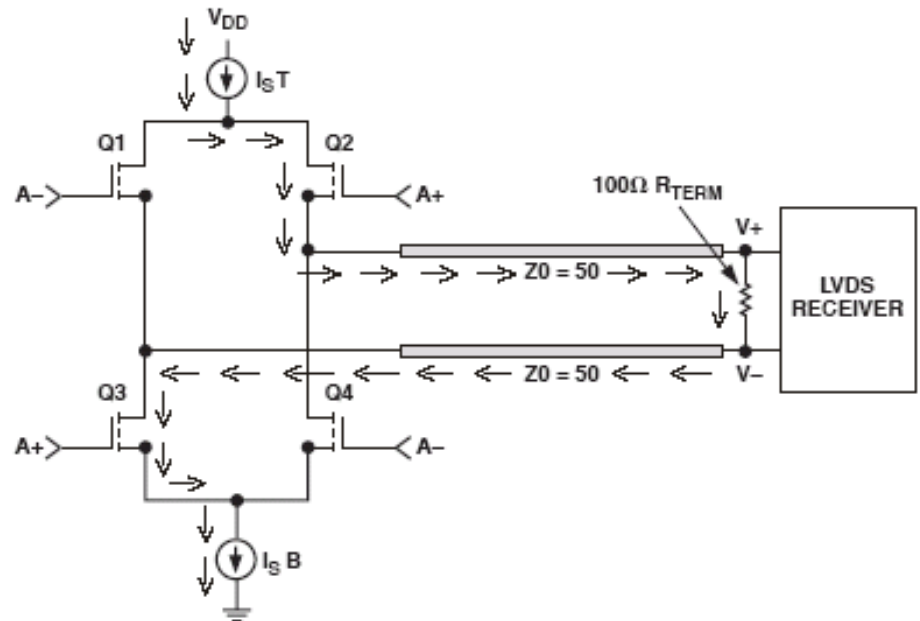


Figure 22. TSSOP Evaluation Board Layout, Secondary Side

ADC Digital Output Signaling

- CMOS signaling works well for low $f_{CLK} < 100\text{MHz}$
- For high frequencies signaling, we prefer LVDS
 - Low voltage differential signaling
- LVDS vs. CMOS:
 - Higher speed 😊
 - More power efficient 😊
 - Less noise 😊
 - Lower swing 😞
 - Two pins/bit 😞

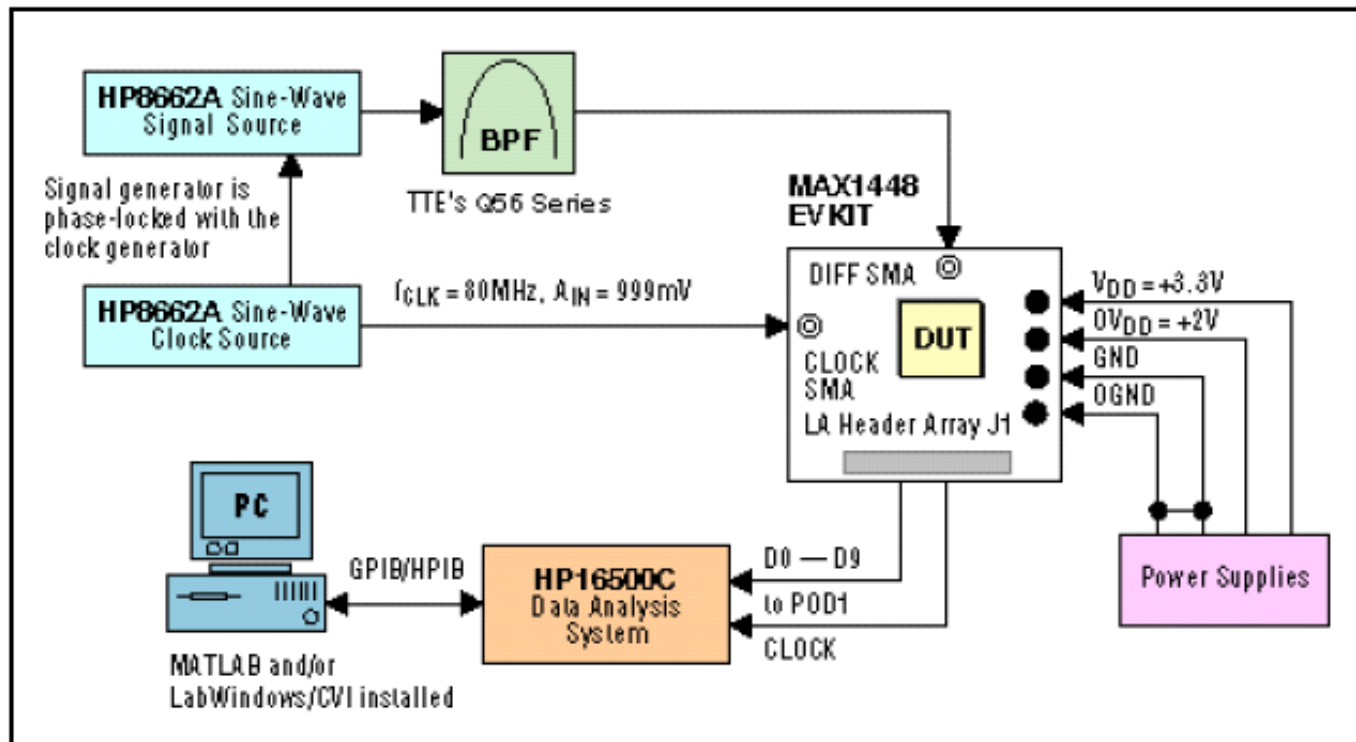


Analog Devices application note 586: "LVDS data outputs for high speed ADCs"

How to Acquire ADC Output

- Several options:
 - Logic analyzer with PC interface
 - FIFO board, interface to PC DAQ card
 - Oscilloscope
- We can build on-chip DACs and send multiple bits through one pin
 - It reduces the pin count 😊
 - It lowers the power 😊
- We can use an oscilloscope to capture the DAC outputs and re-quantize it to obtain the ADC digital output
 - This works especially well for ultra-high-speed ADCs
 - Logic analyzers usually cannot work beyond 2GS/s
 - High-speed (e.g., 50GS/s) oscilloscopes are easy to find

Complete Setup

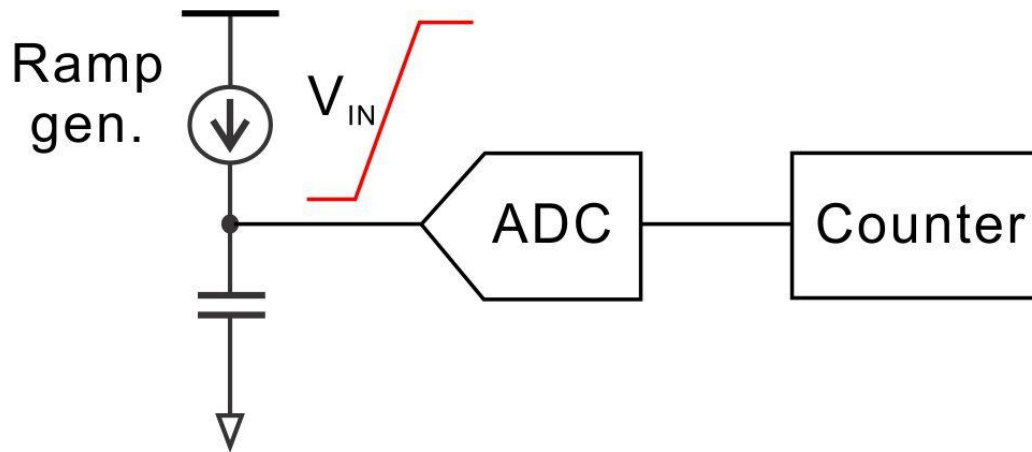


[Maxim Application Note 729: "Dynamic Testing of High-Speed ADCs, Part 2]

Spectral and DNL/INL

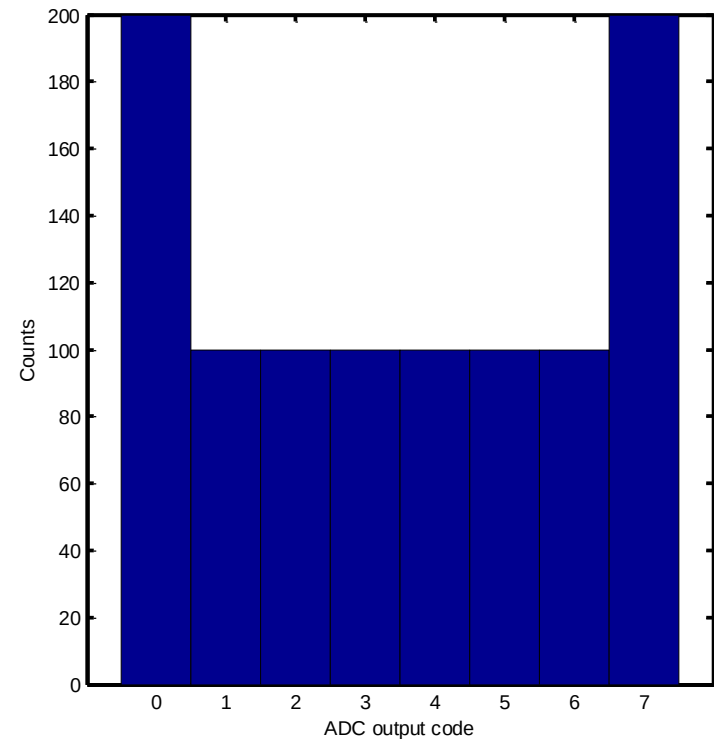
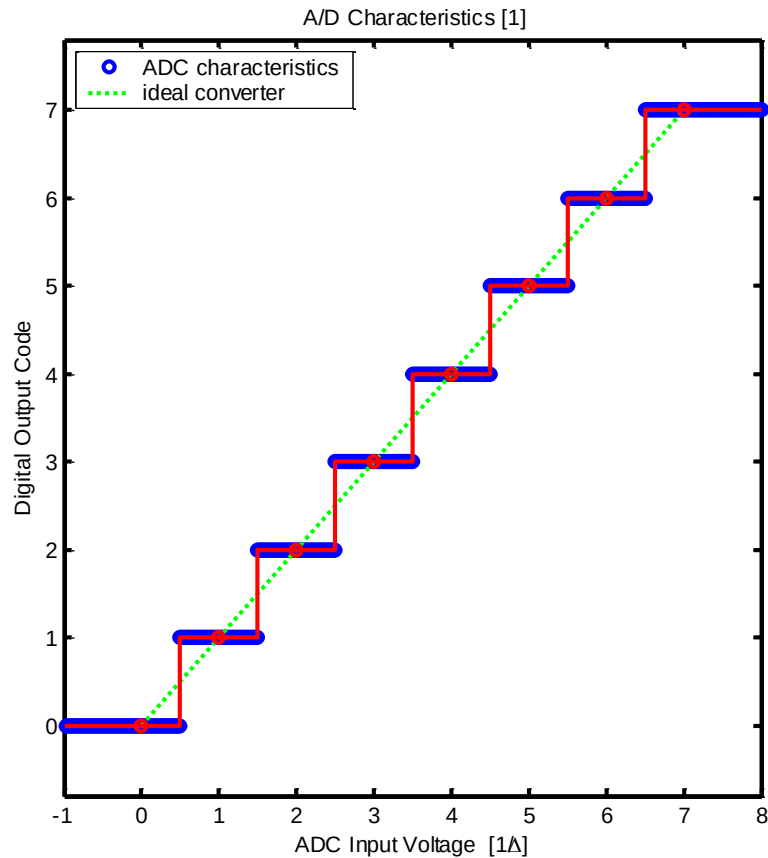
- We can perform FFT in Matlab to measure spectral metrics
- How about DNL/INL for a DAC?
 - "Trivial", apply codes and use "a good voltmeter" to measure outputs
- How about DNL/INL for an ADC?
 - Need to find "decision levels", i.e. input voltages at all code boundaries
 - We can measure code widths through histogram testing

Histogram Test

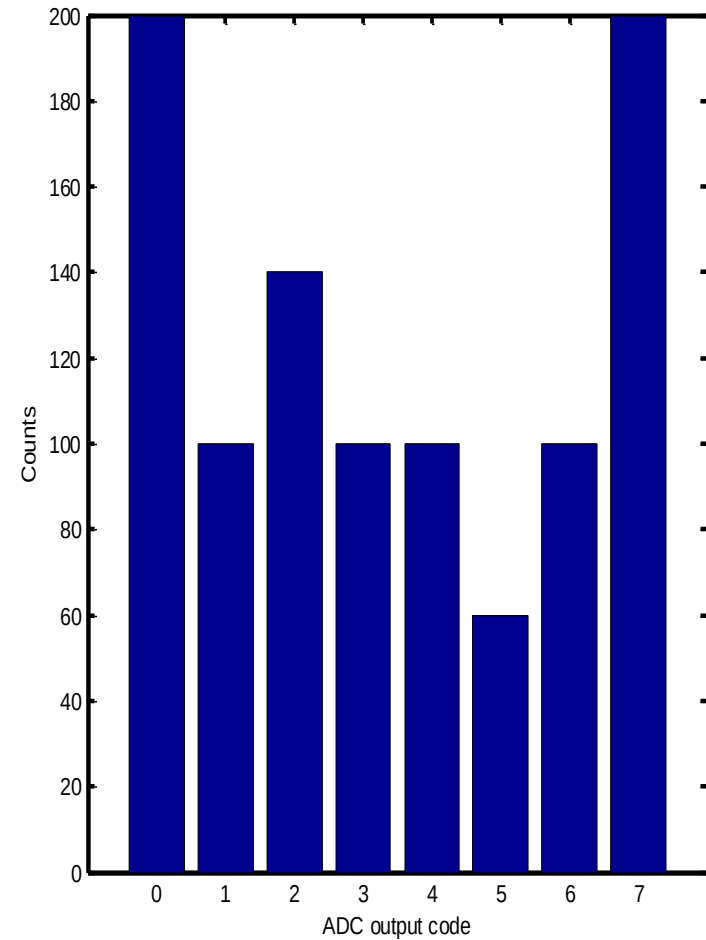
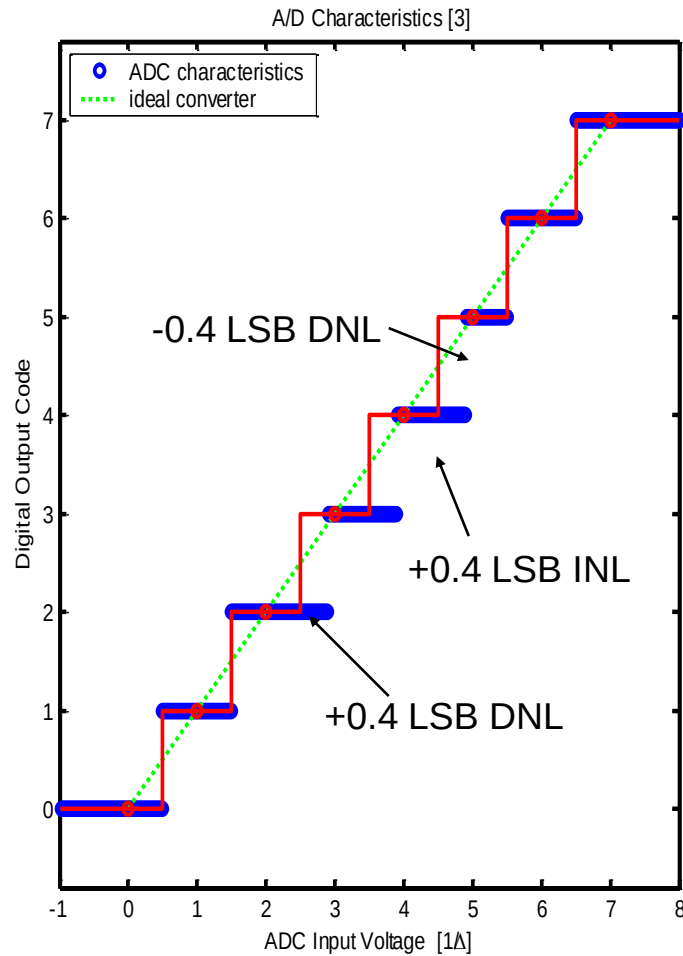


- Code width is proportional to the number of code occurrences
- Ramp needs to be sufficiently slow to provide enough code occurrence
 - For a 16-bit 100 kS/s ADC, to provide on average 100 occurrence per code, we need the ramp period to be $2^{16} \times 100 / 100k \approx 64$ seconds

Histogram of an Ideal 3-Bit ADC

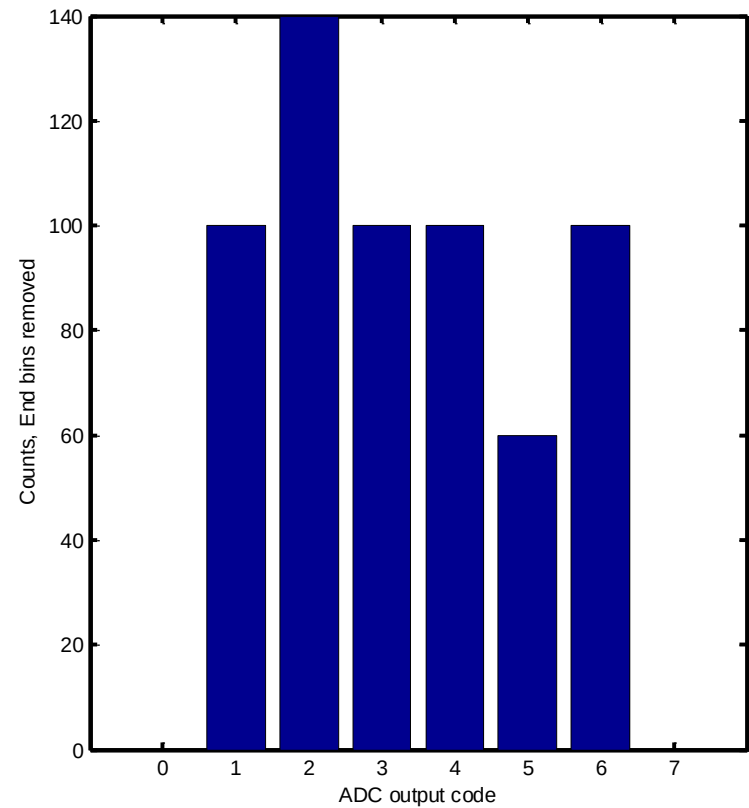


Histogram of a Non-ideal 3-Bit ADC



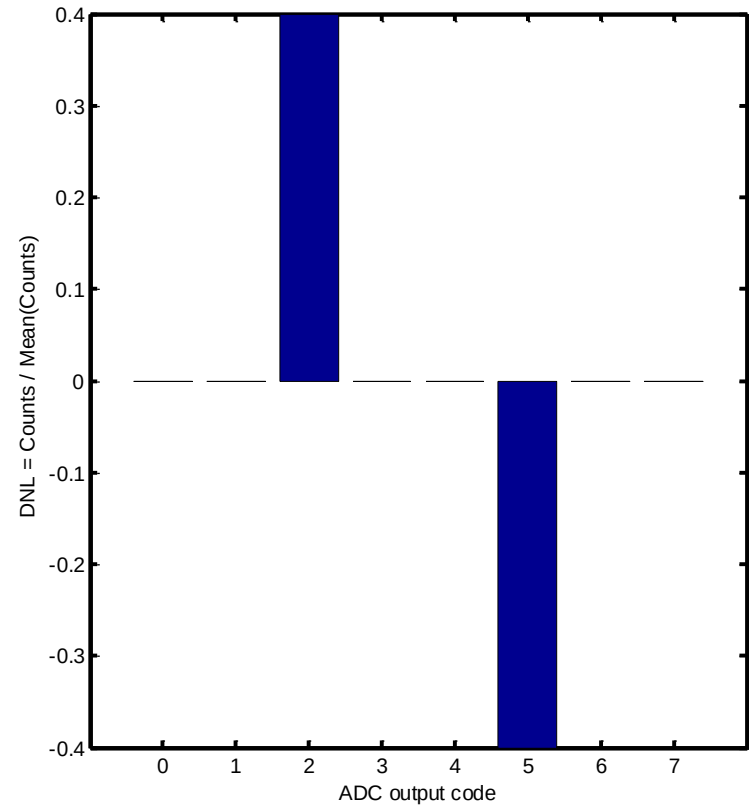
DNL from Histogram (1)

- Step 1
 - Remove “over-range bins” (0 and 7)



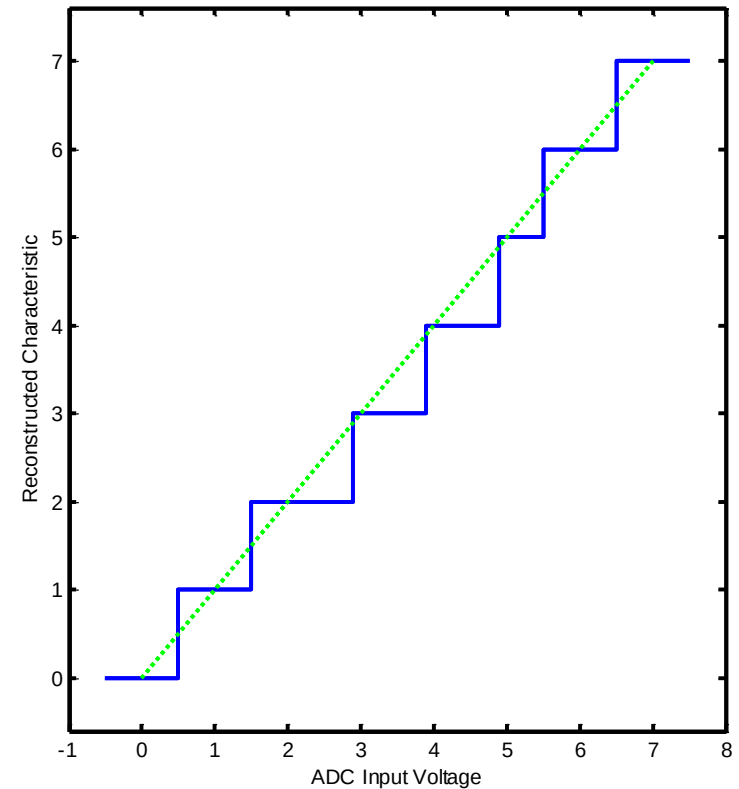
DNL from Histogram (2)

- Step 2
 - Divide by average count
- Step 3
 - Subtract 1
 - Ideal bins have exactly the average count, which corresponds to 1 after normalization
- Result is DNL

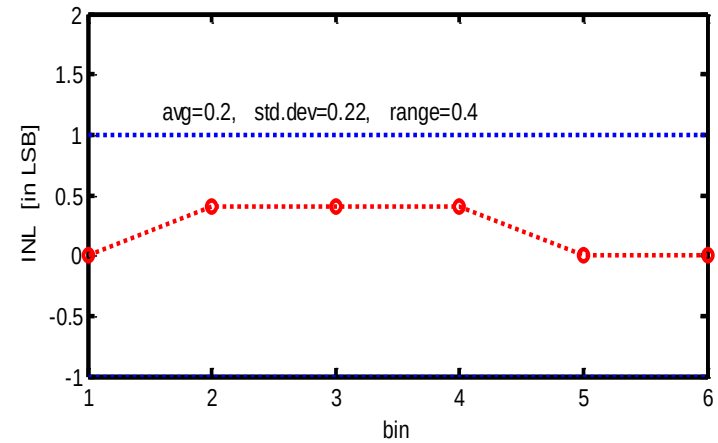
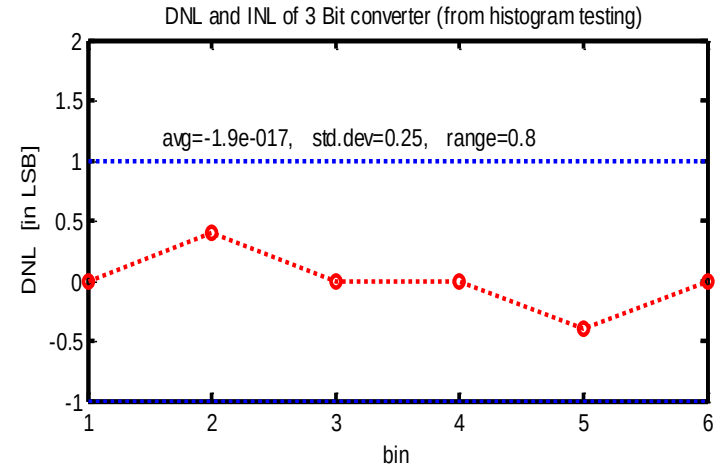
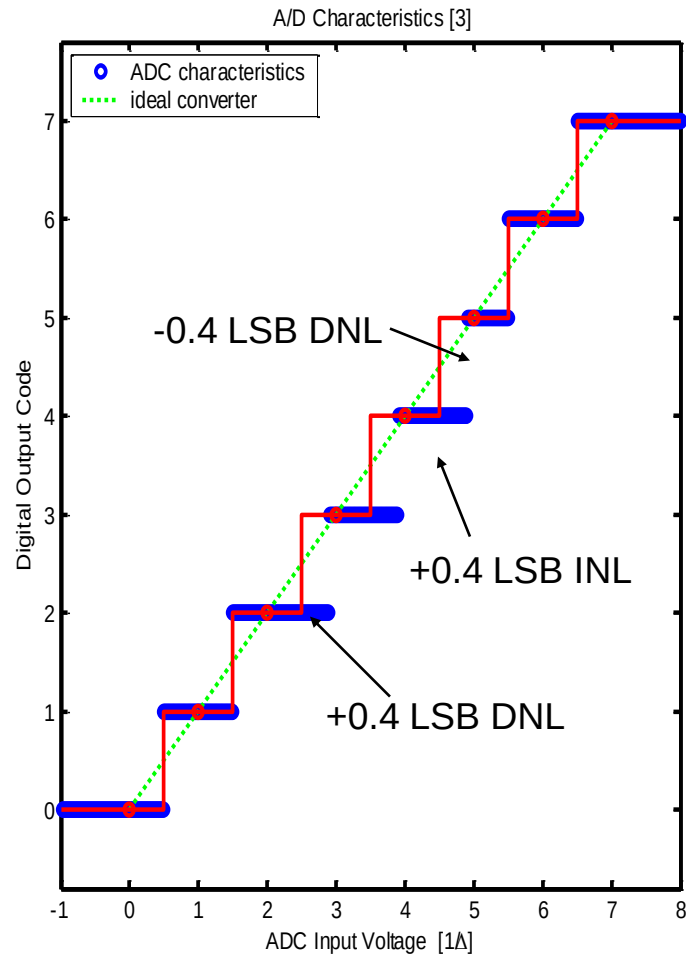


INL from Histogram

- INL is simply running sum of DNL
- The DNL information can also be used directly to construct the converter transfer function
 - Simply add up all bin-widths to find transition levels

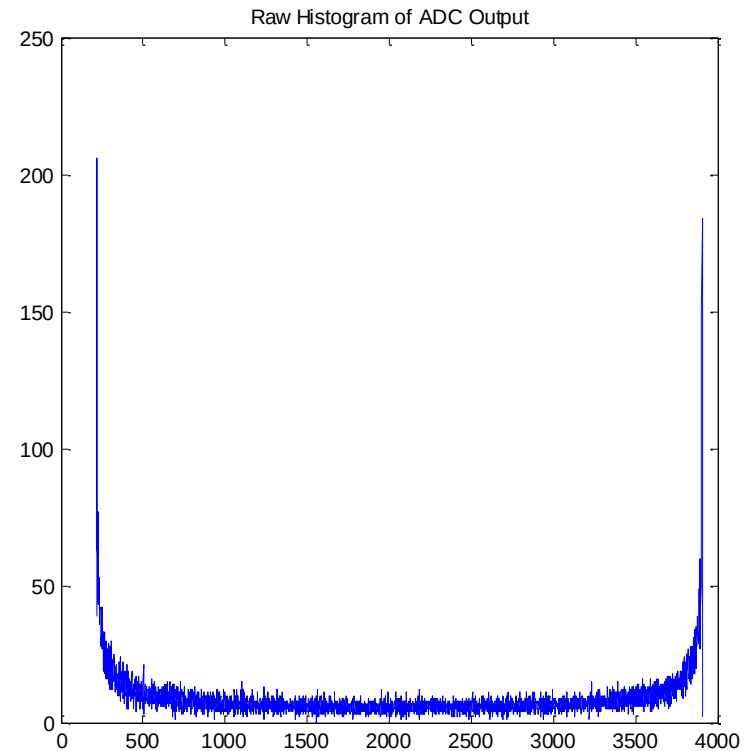


DNL and INL of the ADC

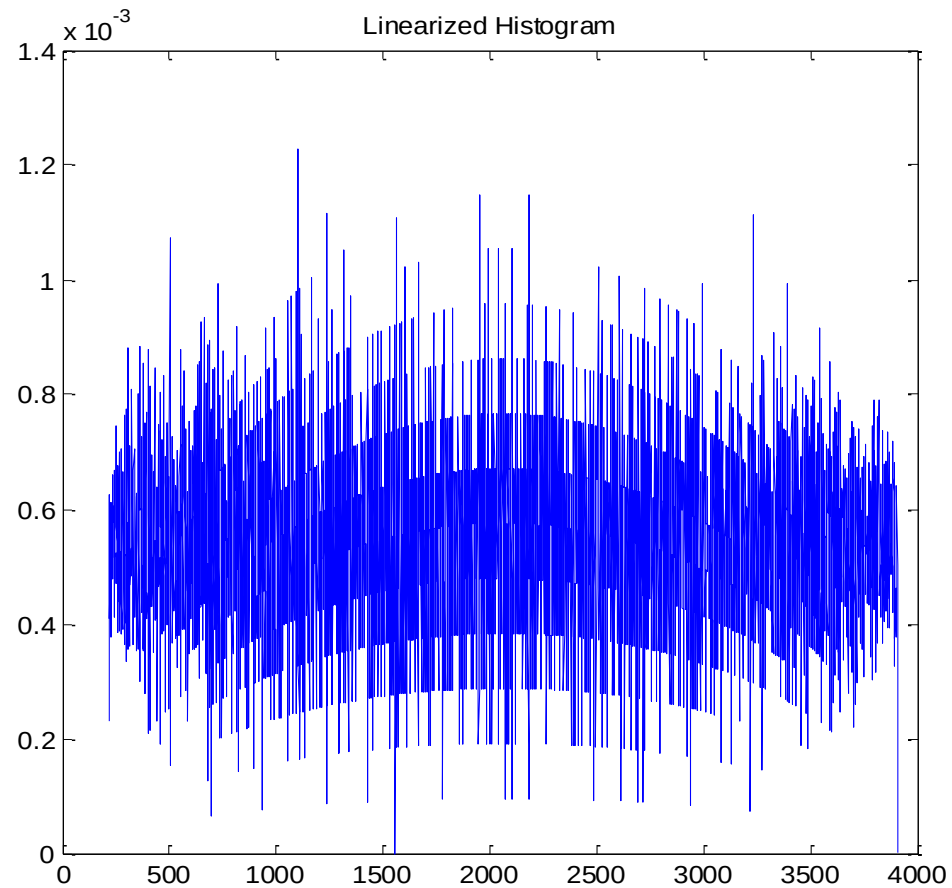


Histogram Test Based on Sine Inputs

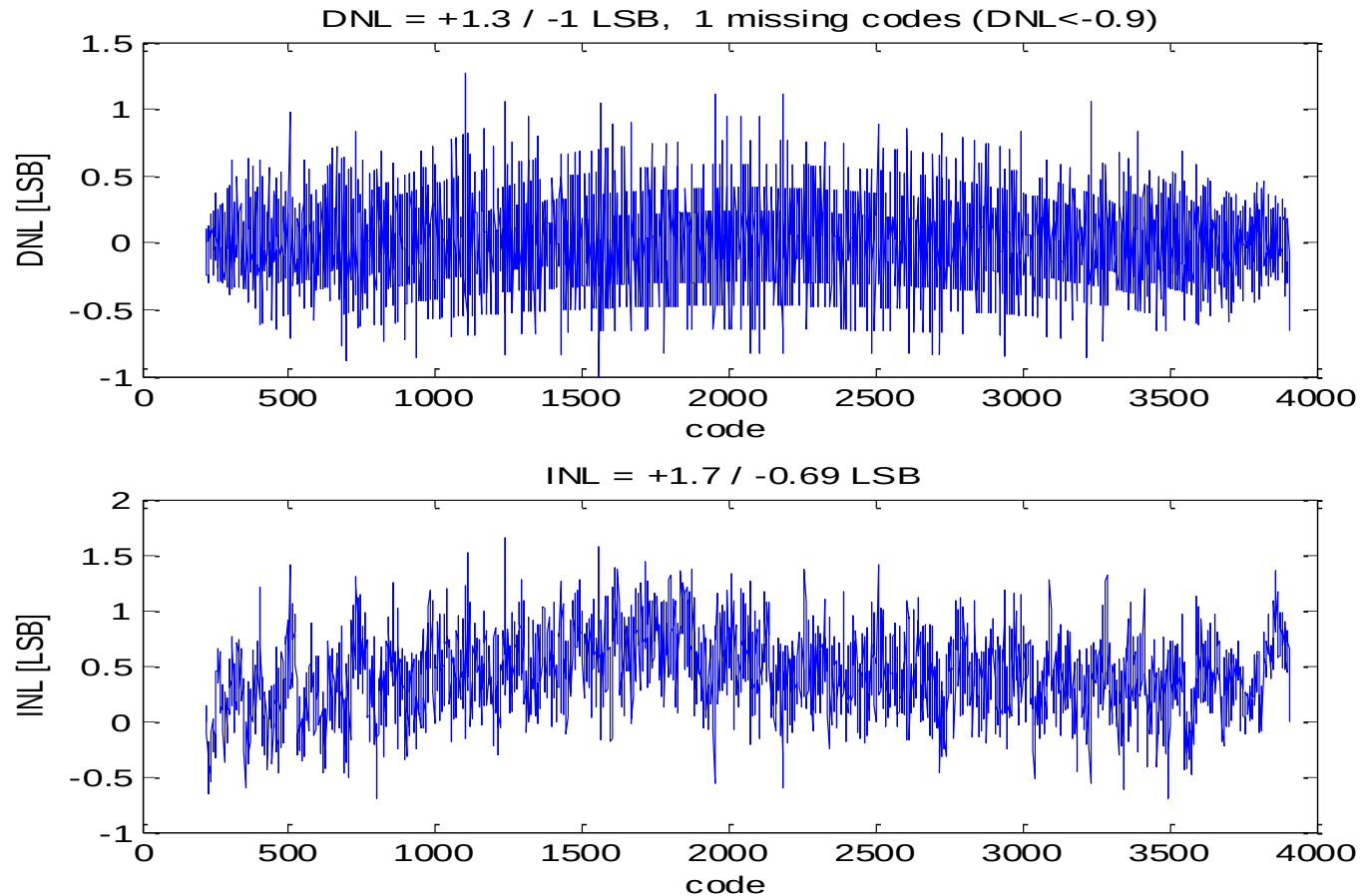
- It is nontrivial to generate highly linear ramps
 - The ramp needs to be more linear than the ADC
 - E.g., for a 16-bit ADC, the ramp needs to be 18-b linear
- It would be nice to use a sinusoidal input to perform histogram test
 - Nature allows perfect sine
- Problem for using a sine input
 - The histogram is not flat
 - We can apply an inversion function



After Correction for Sinusoidal PDF



Resulting DNL and INL



Correction for Sinusoidal PDF

- References
 - M. V. Bossche, J. Schoukens, and J. Renneboog, “Dynamic Testing and Diagnostics of A/D Converters,” IEEE TCAS, Aug. 1986.
 - IEEE Standard 1057
- Do we need to know the exact amplitude and offset of the sine input?
 - No!
 - DNL and INL are independent from amplitude and offset

DNL/INL Code

```
function [dnl,inl] = dnl_inl_sin(y);  
%DNL_INL_SIN  
% dnl and inl ADC output  
% input y contains the ADC output  
% vector obtained from quantizing a  
% sinusoid  
  
% Boris Murmann, Aug 2002  
% Bernhard Boser, Sept 2002  
  
% histogram boundaries  
minbin=min(y);  
maxbin=max(y);  
  
% histogram  
h = hist(y, minbin:maxbin);  
  
% cumulative histogram  
ch = cumsum(h);  
  
% transition levels  
T = -cos(pi*ch/sum(h));  
  
% linearized histogram  
hlin = T(2:end) - T(1:end-1);  
  
% truncate at least first and last  
% bin, more if input did not clip ADC  
trunc=2;  
hlin_trunc = hlin(1+trunc:end-trunc);  
  
% calculate lsb size and dnl  
lsb= sum(hlin_trunc) / (length(hlin_trunc));  
dnl= [0 hlin_trunc/lsb-1];  
misscodes = length(find(dnl<-0.9));  
  
% calculate inl  
inl= cumsum(dnl);
```

DNL/INL Code Test

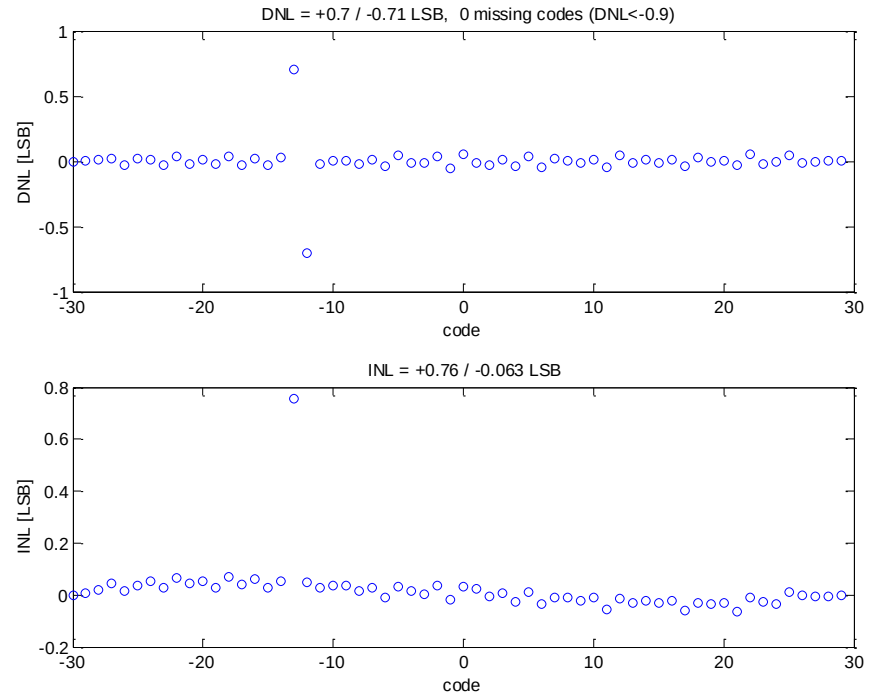
```
% converter model

B = 6; % bits
range = 2^(B-1) - 1;
% thresholds (ideal converter)
th = -range:range; % ideal thresholds
th(20) = th(20)+0.7; % error

fs = 1e6;
fx = 494e3 + pi; % try fs/10!
C = round(100 * 2^B / (fs / fx));

t = 0:1/fs:C/fx;
x = (range+1) * sin(2*pi*fx.*t);
y = adc(x, th) - 2^(B-1);

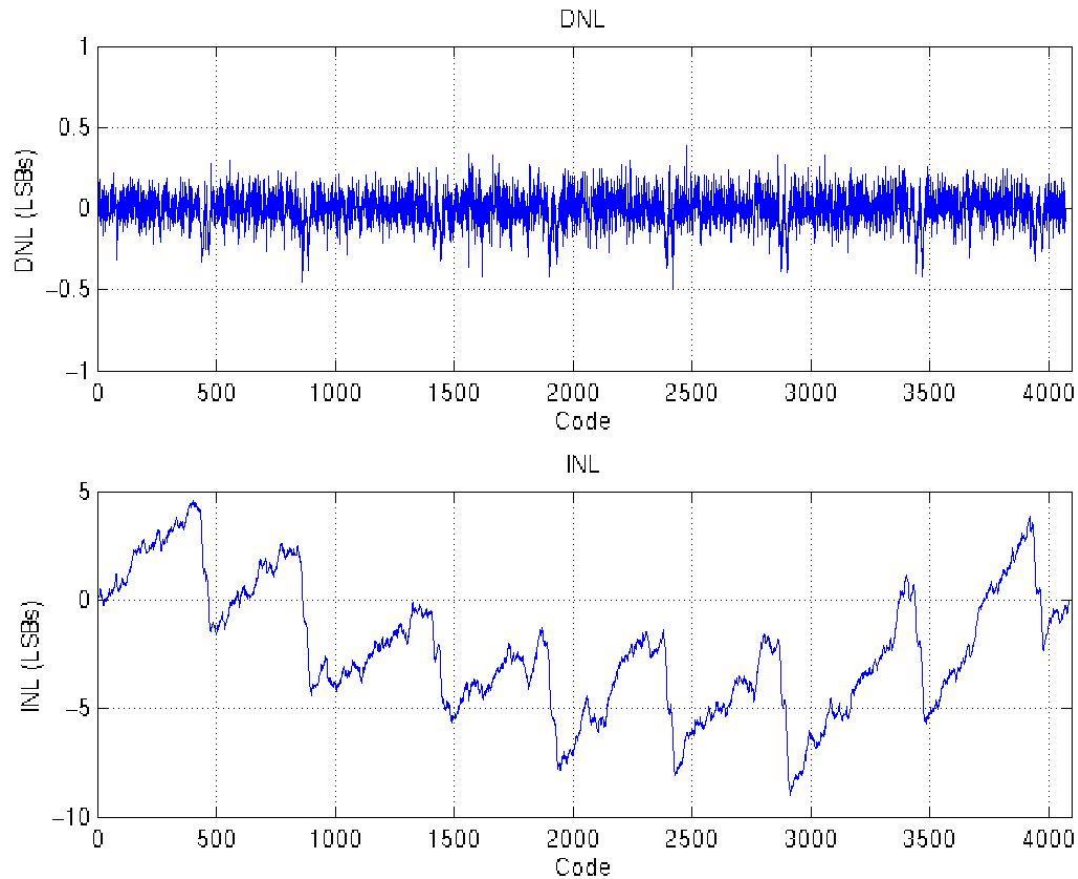
hist(y, min(y):max(y));
dnl_inl_sin(y);
```



Limitations of Histogram Testing

- Histogram testing assumes monotonicity
 - E.g. “code flips” will not be detected
- Dynamic sparkle codes produce only minor DNL/INL errors
 - E.g. 123, 123, ..., 123, 0, 124, 124, ...
 - Must look directly at ADC output to detect
- Noise can smear DNL
 - E.g. 9, 9, 9, 10, 9, 9, 9, 10, 9, 10, 10, 10, ...
 - INL is more robust against noise, need to look at both INL and DNL
- Reference
 - B. Ginetti and P. Jespers, “Reliability of Code Density Test for High Resolution ADCs,” Electron. Letters, pp. 2231-2233, Nov. 1991.

DNL Smeared By Noise



[Source: David Robertson, Analog Devices]